5.7 Septic Tank

Septic tank is a rectangular water tight under-ground masonry tank. The sewage is admitted into it for treatment. It is suitable for disposing the sewage from isolated buildings, institutions, hospitals, hotels, etc.

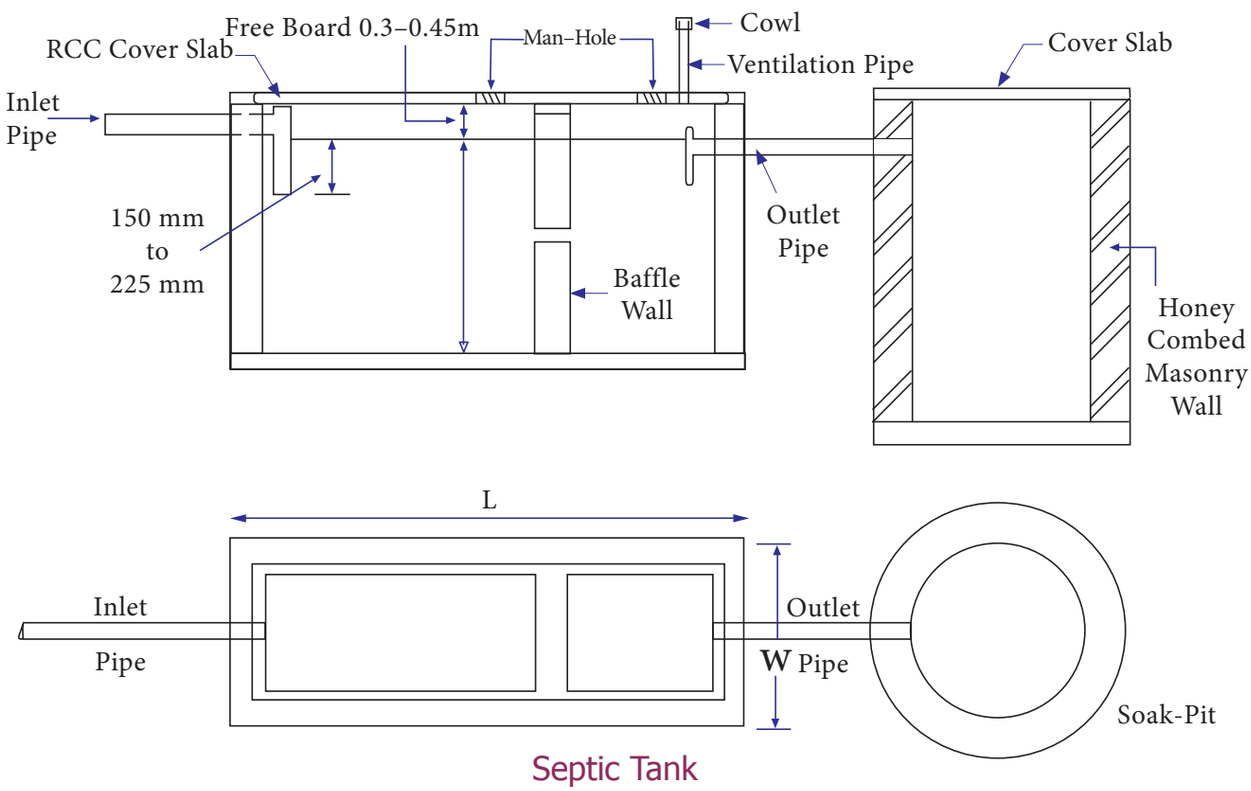
5.7.1 Construction and Working of Septic tank

There are two stages of purification in septic tank. They are,

1. Anaerobic digestion
2. Aerobic Oxidation

Construction Details

1. The tank is rectangular in plan and the length is usually 2 to 4 times the breadth, Normally thrice.
2. Liquid depth of 1 to 2 m and free-board of 0.3 to 0.45 m are provided.
3. An elbow pipe (or) 'T' shaped pipe submerged to a depth of 150 to 225 mm below the liquid level is provided as inlet pipe.
4. T shaped outlet pipe is provided to a depth of 150 mm below the liquid level.
5. Inlet baffle is placed at 1/5th the length of tank from the inlet pipe. In larger tank, the baffle walls are constructed to guide the flow and prevent short circuiting.
6. The tank is divided into two compartments. The first one serves as the stilling compartment and the second one serves as the settling compartment with a smooth exit of settled sewage.
7. The floor is provided with all sides sloping (1 in 7) towards one point to facilitate desludging operation.
8. R.C.C roof slab with a manhole cover in each compartment is provided to permit inspection and maintenance.
9. Ventilation is provided by a pipe 40 to 50 mm in diameter. It is taken well above the roof level.
10. A cowl is provided at the top of the vent pipe. It prevents the birds from nesting.



Septic Tank

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